



GlucoFlow

Smart Diabetes Monitoring & Automated AI Healthcare Support

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[DEMO LINK OF THE HACKATHON](#)

What is the problem?

Global Health crisis

- 200 million in 1990 and 830 million in 2022 globally diabetes cases type 1 and type 2.
- Growth of 315%
- Unmanaged blood sugar causes severe complications, including heart attacks, kidney failure, and neuropathy.
- High blood glucose was also associated with approximately 11% of cardiovascular deaths.

Living with diabetes

- Measuring 4 - 7+ times daily.
- Watching your daily diet.
- Mental health -> anxiety, depression and stress.
- Practical annoyances: sore or calloused fingertips and fatigue with the routine
- "Diabetes is widespread, growing, and many people remain unaware of their condition."

Who is our user group?

- (Newly) diagnosed diabetes patients (type 1 and type 2)
- Caregivers
- Attending clinicians.



What is our solution?

1. Web Application

A simple website where users enter their health information and send it to n8n.

2. n8n Automation

n8n receives the information automatically and starts the process.

3. Gemini AI

Gemini checks the glucose level, symptoms, and other information. It gives one of three results: **Stable, Warning, or Urgent.**

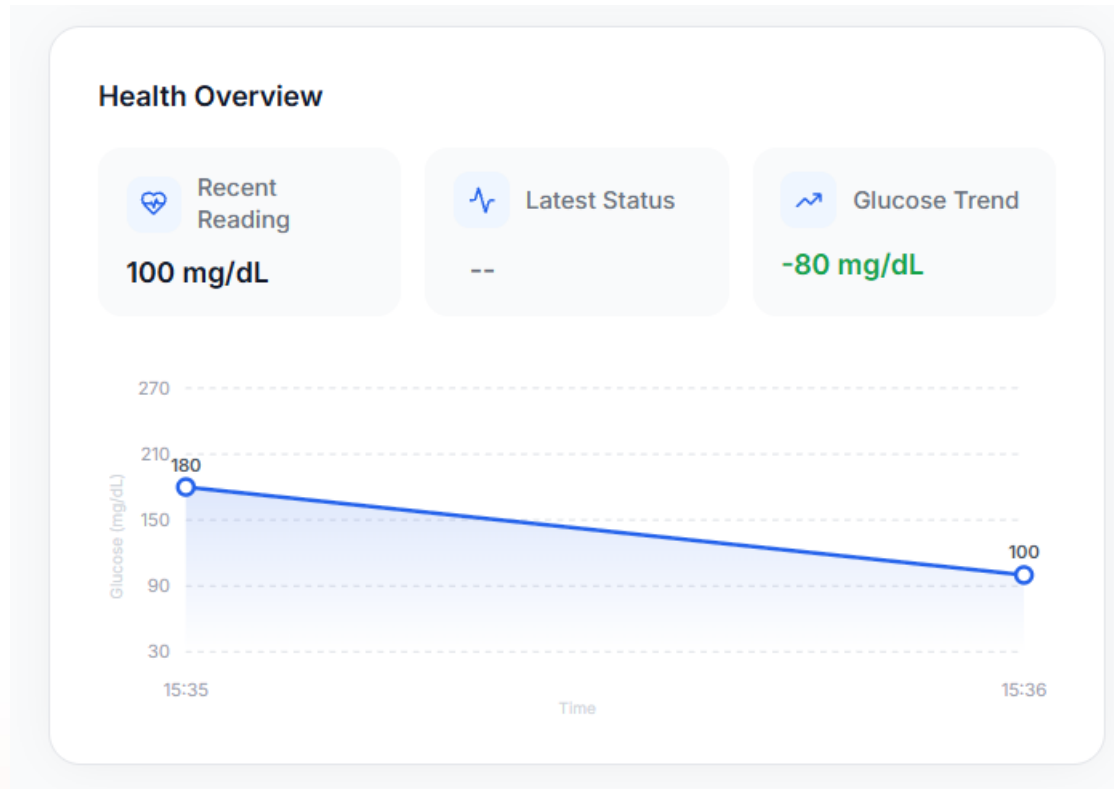
4. Caregiver Alert

If the result is urgent, the system sends a Telegram message to the caregiver.



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Demonstration of the product



Important Test: A glucose level of 300 mg/dL gives an urgent health status.

Emergency Question: The app asks the user if they want to contact a doctor.

Telegram Alert: The caregiver gets an 🚨 URGENT message with the health information.

Glucose Graph: The user can see their glucose levels over time.

Ethical Reflection & Safety

Automation Boundary: AI is used strictly for contextualization and triage alerts, never for prescribing medical treatment or insulin dosages.

User's countries: Different parts of the world have different type of care or testing for diabetes. Most people do not know they have diabetes.

Data Privacy & Security: Built with header-ready webhook parameters to ensure HIPAA-compliant patient data encryption in production.

Financial difference: People with fewer income who can not pay for a phone or medical health care, unfortunately can not use our product.

Questions?

